HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME		**CT GROUP**	**13S** ____
CENTRE NUMBER		**INDEX NUMBER**	

BIOLOGY **9648/01**

Paper 1 Multiple Choice **24 September 2014**

Additional Materials: Optical Mark Sheet **1 hour 15 minutes**

INSTRUCTIONS TO CANDIDATES

1. Write your **name**, **CT group**, **centre number** and **index number** in the spaces provided at the top of this cover page.
2. Fill in your particulars on the Optical Mark Sheet. Write your **NRIC number** and shade accordingly.
3. There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D.**

   Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.
4. At the end of the paper, you are to submit **only** the Optical Mark Sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

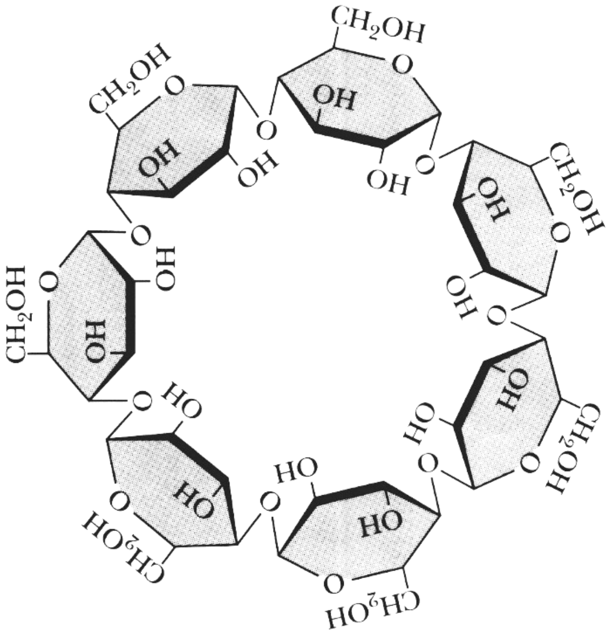
1 The distribution of membranes in the organelles of three cell types, a prokaryotic cell, a liver cell and an enzyme-secreting cell from the stomach, is determined. For each organelle, the amount of membrane is expressed as a percentage of the total amount of membrane in the cell.

Which row of values best represents the membrane distribution in the three cell types?

CM : cell membrane
rERM : rough endoplasmic reticulum membrane
IMM : inner mitochondrial membrane
OMM : outer mitochondrial membrane

	prokaryotic cell			liver cell			stomach cell		
	CM	rERM	IMM	rERM	OMM	IMM	rERM	OMM	IMM
A	99	0	0	35	7	28	60	16	4
B	99	0	0	35	7	28	60	4	16
C	52	12	36	60	7	28	35	4	16
D	52	12	36	60	28	7	35	4	16

2 The diagram shows a circular oligosaccharide molecule.



In which other molecule can a similar glycosidic bond be found?

A lactose

B maltose

C sucrose

D cellulose

3 The hydrolysis of triglycerides leads to

1 the formation of products which are more soluble in water than triglycerides.
2 the formation of products which are less soluble in water than triglycerides.
3 an increase in pH.
4 a decrease in pH.

A 1 and 4

B 2 and 3

C 2 and 4

D 1 and 3

4 In an investigation, a solution of enzyme **X** is placed in a sealed bag made of selectively permeable membrane. The bag is then placed in a beaker of distilled water, and the distilled water is changed several times.

Samples of equal volume are then drawn from the bag at regular intervals. To each sample extracted, a fixed amount of substrate was added and the rate of reaction is determined by the number of bubbles formed over 10 minutes. The number of bubbles formed is observed to be inversely proportional to the duration the bag is immersed in distilled water.

Which is a valid conclusion that can be made from this investigation?

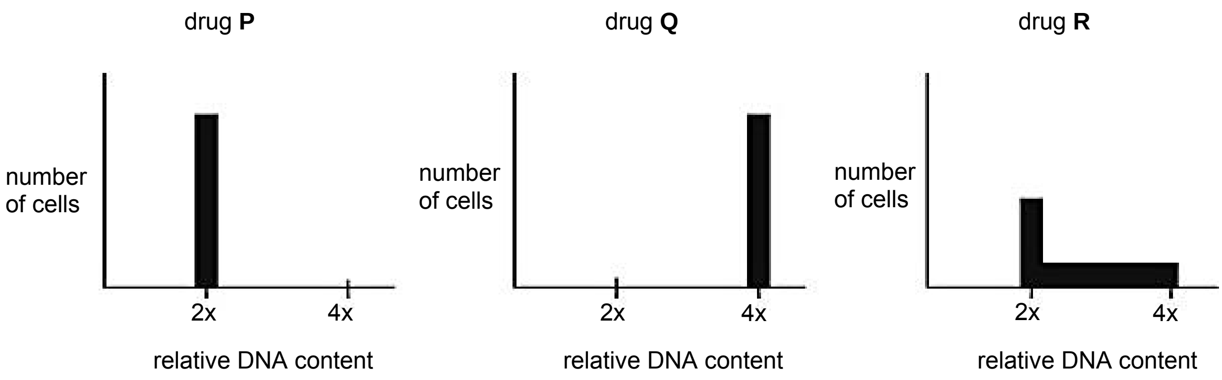
A Some enzyme **X** molecules have lost their secondary and tertiary structure after some time.

B More enzyme **X** molecules were inhibited by a non-competitive inhibitor present in the solution over time.

C Concentration of enzyme **X** decreases with time.

D Enzyme **X** is moving out of the bag at a slow rate.

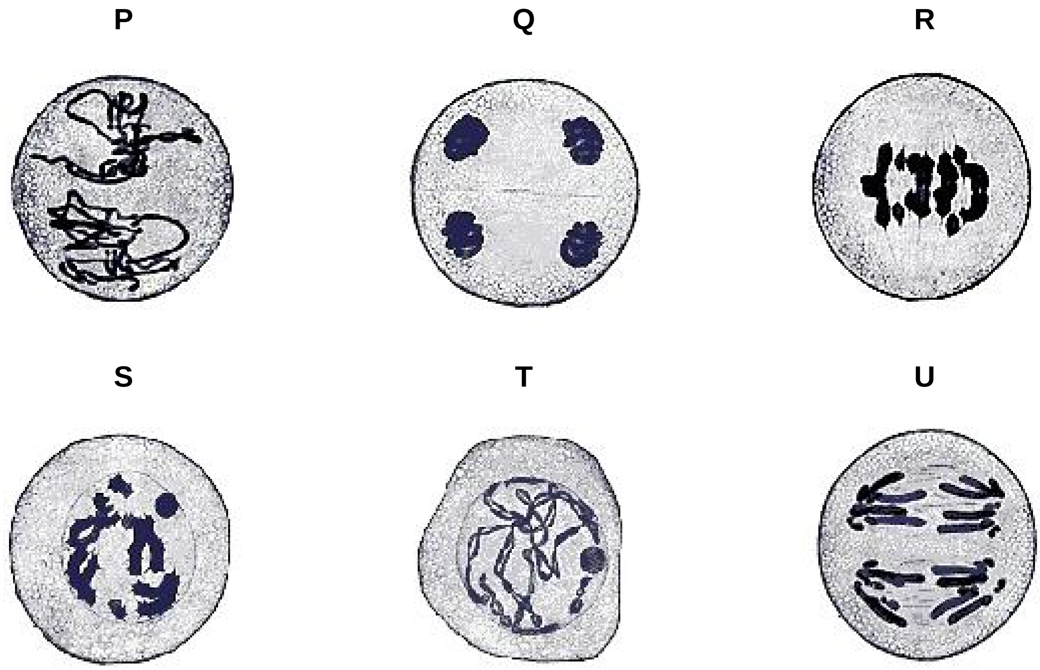
5 Three batches of cells at the start of G1 phase of the cell cycle were treated with three drugs **P**, **Q** and **R**. Each batch has the same number of cells. Untreated cells are expected to complete interphase within 24 hours. The relative DNA content of the treated cells was measured after 24 hours. The three graphs show the results obtained.



What is the effect of each drug on the cells?

	drug **P**	drug **Q**	drug **R**
A	blocks G1 phase	blocks S phase	blocks G2 phase
B	blocks G2 phase	blocks S phase	blocks G1 phase
C	blocks S phase	blocks G1 phase	blocks G2 phase
D	blocks G1 phase	blocks G2 phase	blocks S phase

6 Cells from the aloe vera plant are shown at different stages of meiosis in micrographs **P** to **U**.



Which sequence shows the correct order of these micrographs?

A **P → S → R → T → U → Q**

B **S → T → R → P → Q → U**

C **T → S → R → P → U → Q**

D **T → R → S → Q → P → U**

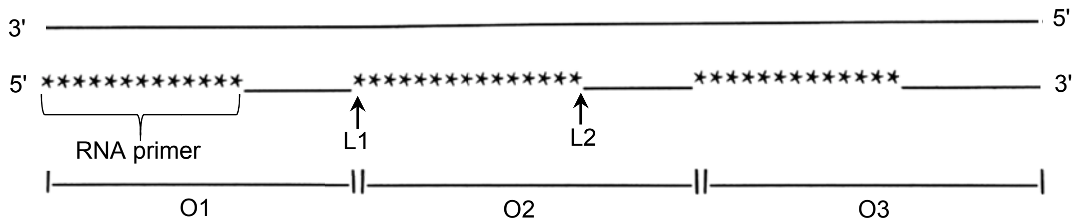
7 A 19-base pair long DNA molecule was analysed to find the number of nucleotide bases in each of the polynucleotide strands. Some of the results are shown.

	number of nucleotide bases			
	A	C	G	T
strand 1				4
strand 2		7		5

How many hydrogen bonds are present in this DNA molecule?

A 31 **B** 48 **C** 39 **D** 57

8 The diagram shows a DNA template with the lagging strand prior to the removal of the RNA primers.



Which row correctly shows the events taking place during the synthesis of the lagging strand?

	first Okazaki fragment synthesised	site of phosphodiester bond formation catalysed by DNA ligase
A	O1	L1
B	O1	L2
C	O3	L1
D	O3	L2

9 A segment of a polypeptide chain is Arg – Gly – Ser – Phe – Val – Asp – Arg. It is encoded by the following segment of DNA:

strand 1 3' G G C T A G C T G C T T C C T T G G G G A 5'
strand 2 5' C C G A T C G A C G A A G G A A C C C C T 3'

The genetic code

	U	C	A	G	
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	STOP	STOP	A
	Leu	Ser	STOP	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

Using the genetic code, which row is correct?

	template strand	mRNA codon for Arg
A	1	AGG
B	1	GGA
C	2	GGC
D	2	AGG

10 The following statements describe gene mutation.

1 It can occur in both somatic and sex cells.
2 It can cause Down's syndrome in humans.
3 It can change a dominant allele into a recessive allele.
4 It can increase the number of base pairs in a gene.

Which statement(s) is/are **not** correct?

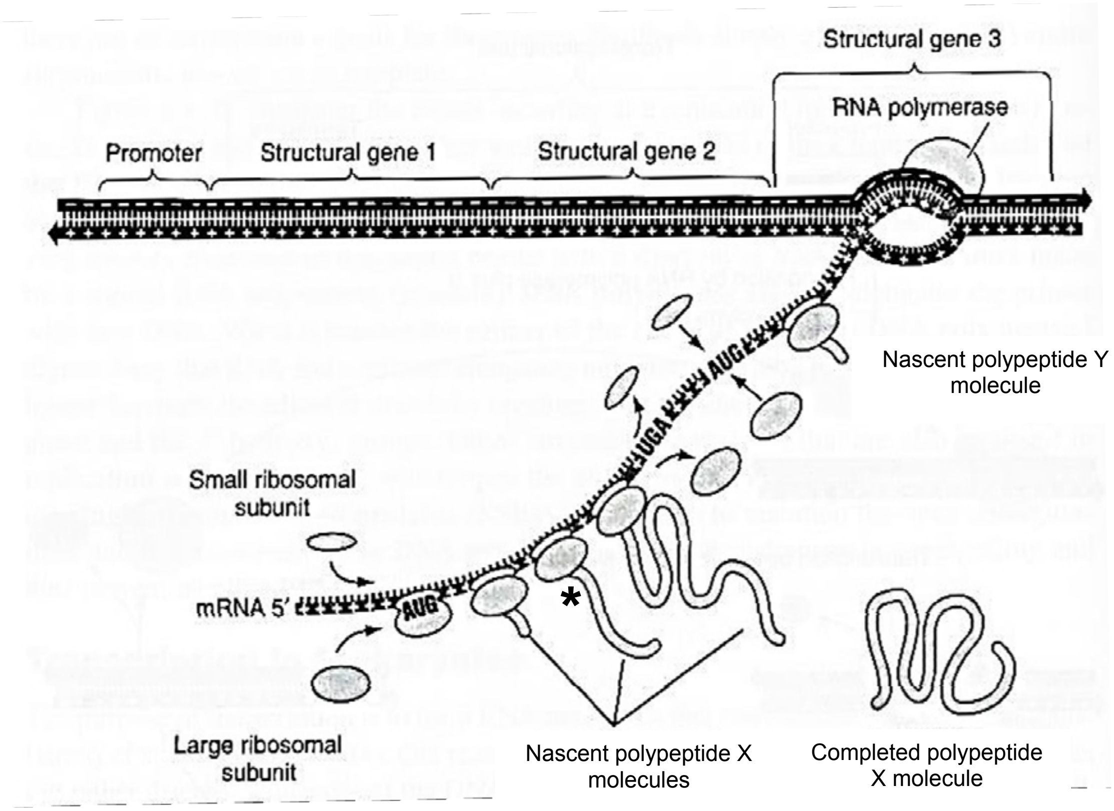
A 2 only

B 3 only

C 2 and 4

D 1, 3 and 4

11 The diagram shows gene expression in an organism.



Which statements are correct?

1 Structural gene 1 encodes protein **X**.
2 The first polypeptide **X** is always synthesised before the first polypeptide **Y**.
3 Synthesis of the polycistronic mRNA allows transcription and translation to be more efficient in this eukaryotic cell.
4 The asterisk (*) indicates the position of the first amino acid, methionine, in polypeptide **X**.

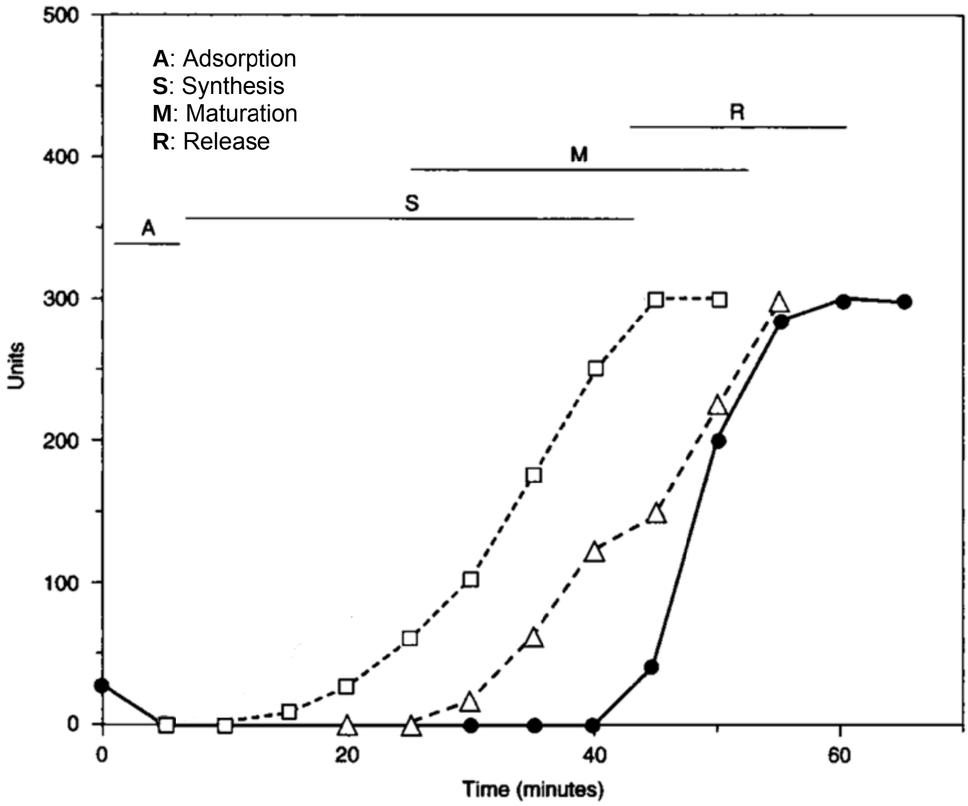
A 1 and 2

B 2 and 3

C 1, 2 and 4

D 1, 3 and 4

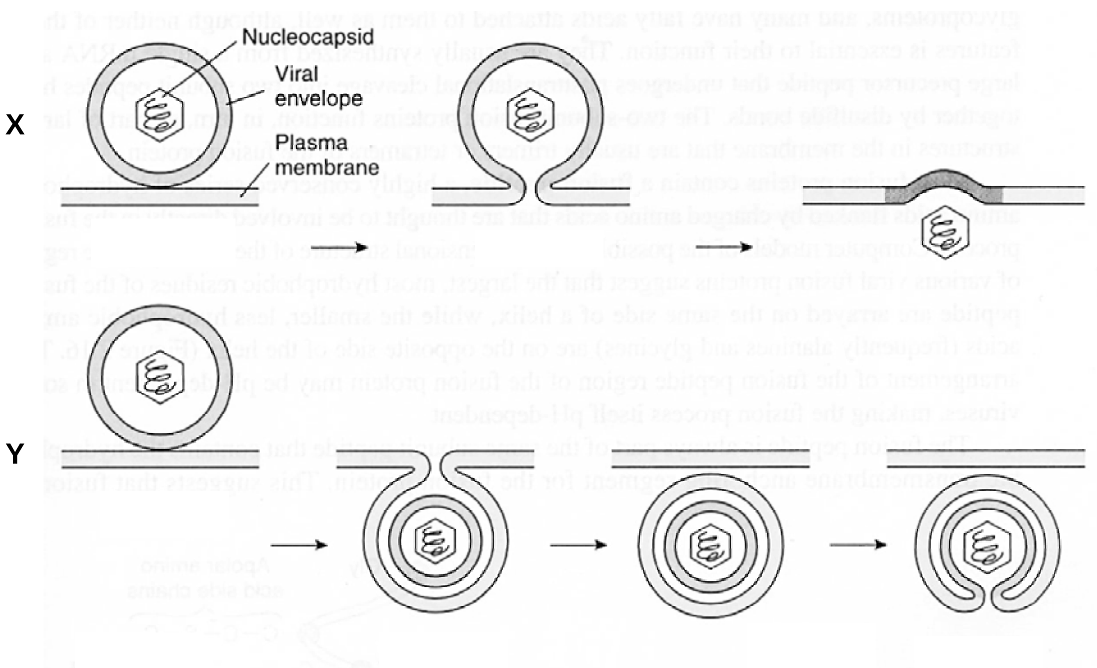
12 The graph represents the process of viral replication.



Which row is correct?

	●	□	△
A	extracellular virions	nucleic acids and proteins	intracellular viral particles
B	extracellular virions	nucleic acids	intracellular viral particles
C	intracellular viral particles	nucleic acids and proteins	extracellular virions
D	intracellular viral particles	nucleic acids	extracellular virions

13 The diagram shows the entry of two types of animal viruses **X** and **Y** into host cells.



Which statements are true about the entry of viruses **X** and **Y**?

1. For **X**, the viral envelope fuses with the plasma membrane of the host cells to allow entry of the virion into the host cell.
2. For **Y**, the viral envelope fuses with the endosomal membrane to release the viral genetic material.
3. For **Y**, the endosomal membrane fuses with the plasma membrane to release the viral genetic material.

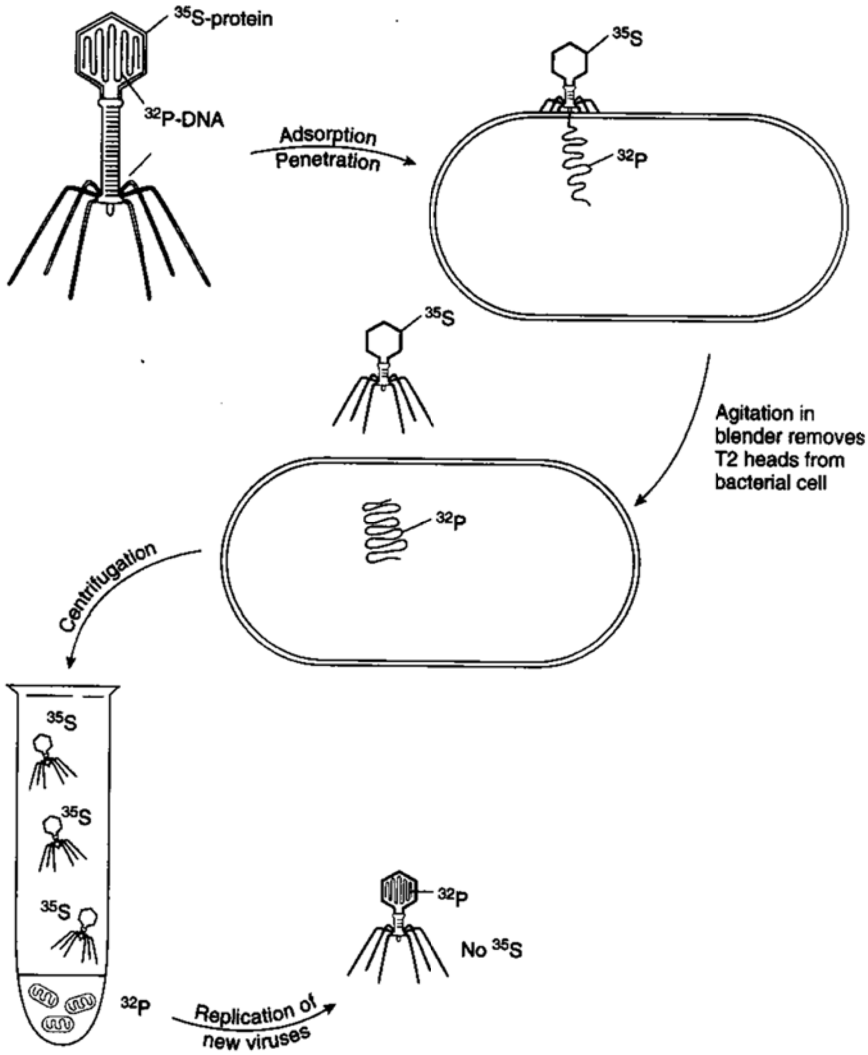
A 1 and 2

B 1 and 3

C 2 and 3

D 1, 2 and 3

14 In the Hershey-Chase experiment, T2 bacteriophages were labelled with radioactive isotopes ^{32}P and ^{35}S in their DNA and proteins respectively. The phages were allowed to infect *Escherichia coli* cells. Then they were subjected to agitation in a blender to remove the T2 heads from the bacterial cells, and centrifuged.



Which statements can be deduced from the experiment?

1 Adsorption to the host cell is required for the reproduction of the T2 bacteriophage.

2 Only the nucleic acid, and not the capsid protein, of the T2 bacteriophage could enter the host cell.

3 After centrifugation, the nucleic acid of the T2 bacteriophage is found in the supernatant.

4 DNA, not protein, is the hereditary material.

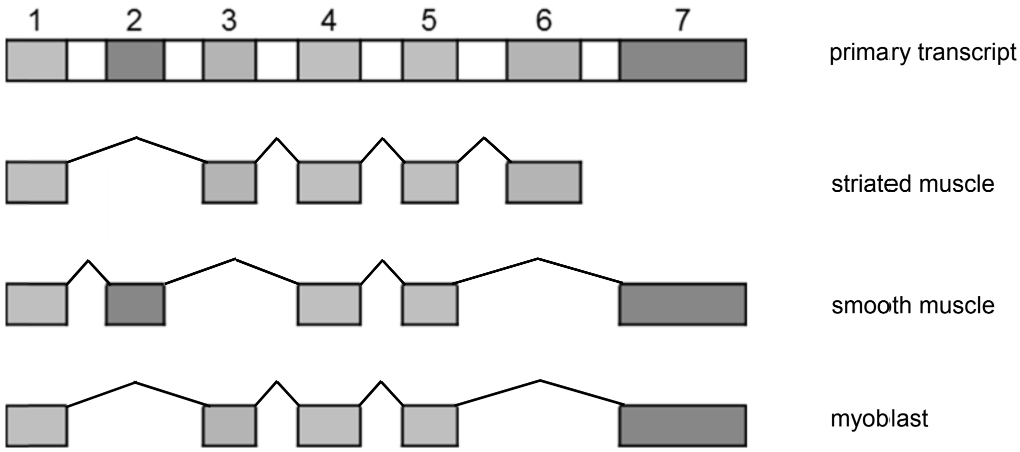
A 1 and 2

B 2 and 4

C 1, 3 and 4

D 1, 2, 3 and 4

15 A single rat gene containing exons numbered 1 to 7 encodes several tissue-specific variants of the muscle protein α-tropomyosin, as shown by the mRNA transcripts in the diagram. The kinked lines indicate the introns and / or exons removed during mRNA splicing in the various muscle tissues.



What cannot be concluded from the diagram?

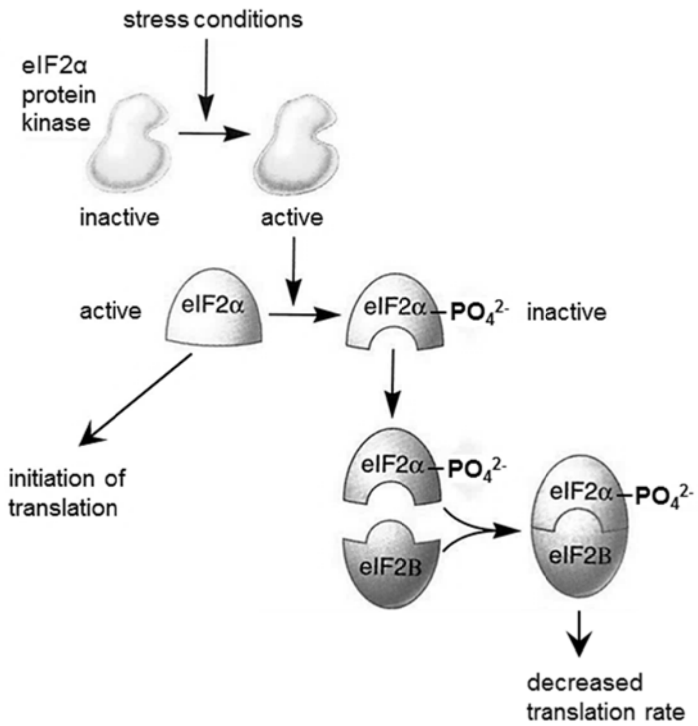
A Exon 1 contains the 5' UTR while both exons 6 and 7 contain the polyadenylation signal.

B Exons 4 and 5 may be important in the function of α-tropomyosin.

C α-tropomyosin expressed in striated muscles has fewer amino acids than those expressed in the smooth muscle and myoblast.

D Deletion of a splice site between exons 1 and 3 will result in non-functional α-tropomyosin in the smooth muscle.

16 The diagram shows a mechanism by which gene expression is controlled during translation.



Which statements are correct?

1 Phosphorylation by eIF2α protein kinase changes the conformation of and activates eIF2α.

2 The concentration of eIF2B affects the rate of translation by inhibiting translation initiation.

3 This regulation results in decreased rate of mRNA translation under stress conditions.

4 The eIF2α-eIF2B complex is recognised by proteasomes for selective degradation due to the presence of the phosphate group.

A 1 and 2

B 1 and 4

C 2 and 3

D 3 and 4

17 Which statements about cancer are correct?

1 Exposure to UV light from the sun always causes skin cancer.
2 Some viruses can cause cancer.
3 The mutation of genes controlling the cell cycle may cause cancer.
4 Inhaling too much carbon monoxide increases the risk of developing cancer.

A 1 and 2

B 1 and 3

C 2 and 3

D 3 and 4

18 In the plant *Antirrhinum*, two genes **A**/**a** and **B**/**b** affect leaf colour.

- The dominant allele, **A**, codes for yellow leaves and is lethal when homozygous.
- The recessive allele, **a**, codes for green leaves.
- The dominant allele, **B**, codes for chlorophyll production, giving green leaves.
- The recessive allele, **b**, results in white leaves and is lethal when homozygous.

When both alleles **A** and **B** are present, the leaves are yellow.

Which will be observed in the offspring produced by two heterozygous plants?

A 50% of the offspring will survive to reproduce.

B 3 out of 16 of the offspring will have white leaves.

C All the offspring with yellow leaves will not survive.

D Ratio of offspring with yellow leaves to offspring with green leaves is 2:1.

19 Two genes involved in determining the coat colour of goats are found on different chromosomes.

The colour gene C causes the hairs to have uniform colour and has three alleles.

- C^{DB} giving dark brown hairs
- C^{B} giving black hairs
- C^{MB} giving medium brown hairs

A dominant allele of the agouti gene (A^{G}) causes the development of white hairs between the coloured hairs giving the coat a shaded appearance.

The table shows the results of crosses between a male goat and 2 female goats.

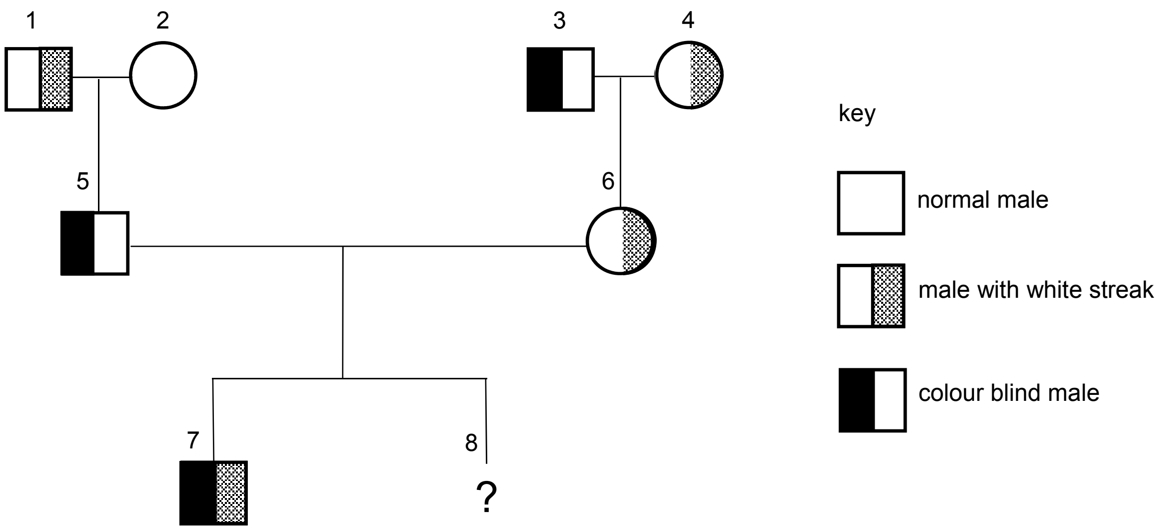
parents	offspring
black agouti male x uniformly dark brown female	50% uniform, 50% agouti; 50% dark brown, 50% black
black agouti male x medium brown agouti female	all agouti; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female parents?

	black agouti male	uniformly dark brown female	medium brown agouti female
A	$C^{B}C^{MB}A^{G}A^{G}$	$C^{B}C^{DB}A^{G}A^{g}$	$C^{MB}C^{MB}A^{G}A^{G}$
B	$C^{B}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{g}A^{g}$	$C^{B}C^{MB}A^{G}A^{G}$
C	$C^{B}C^{DB}A^{G}A^{G}$	$C^{DB}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{G}A^{g}$
D	$C^{B}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{g}A^{g}$	$C^{MB}C^{MB}A^{G}A^{G}$

20 Colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, $\mathbf{X^b}$, is recessive to the allele for normal colour vision, $\mathbf{X^B}$. The gene controlling the presence of a white streak in the hair is not sex-linked, with the allele for the presence of a white streak, **H**, being dominant to the allele for the absence of a white streak, **h**.

The diagram shows a pedigree in which some of the individuals have colour blindness or have a white streak present in the hair.



What is the probability that individual 8 is a male with the same phenotype as individual 7?

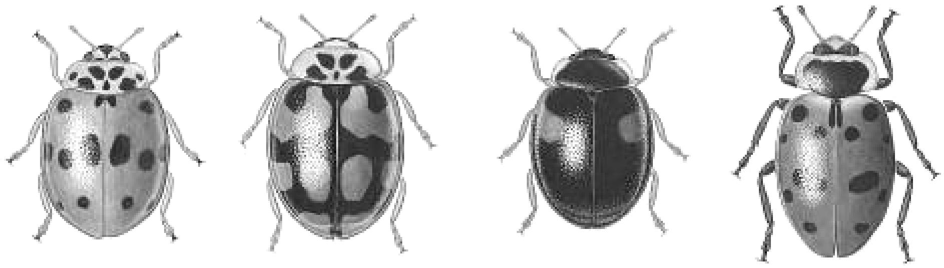
A 0.125

B 0.25

C 0.5

D 0.75

21 An entomologist collected from one habitat four types of black, white and red ladybird beetle (*Coccinellidae*) as illustrated.



Each of the forms were of both sexes and when crossed together with the other colour forms, produced fertile offspring in one or more of the discrete colour forms without intermediates.

What can be concluded from this investigation?

A The beetles are from four closely related species.

B The species has colour forms that are continuously variable.

C The colour forms are influenced by environment and are not under genetic control.

D The species shows a discontinuous variation due to genetic polymorphism.

22 Coat colour in mice is controlled by two genes, each with two alleles. The genes are on different chromosomes.

One gene controls pigment colour. The presence of allele **A** results in a yellow and black banding pattern on individual hairs, producing an overall grey appearance called agouti. Mice with the genotype **aa** do not make the yellow pigment and are, therefore, black.

The other gene determines whether any pigment is produced. The allele **D** is required for development of coat colour. Mice with the genotype **dd** produce no pigment and are called albino.

An albino mouse is mated with a black mouse to produce 12 albino mice, 7 agouti mice and 5 black mice.

A χ^2 test was performed to test the significance of the difference between the observed and expected results.

distribution of χ^2			
number of degrees of freedom (v)	probability		
	0.1	0.05	0.01
1	2.71	3.84	6.64
2	4.60	5.99	9.21
3	6.25	7.82	11.34
4	7.78	9.49	13.28

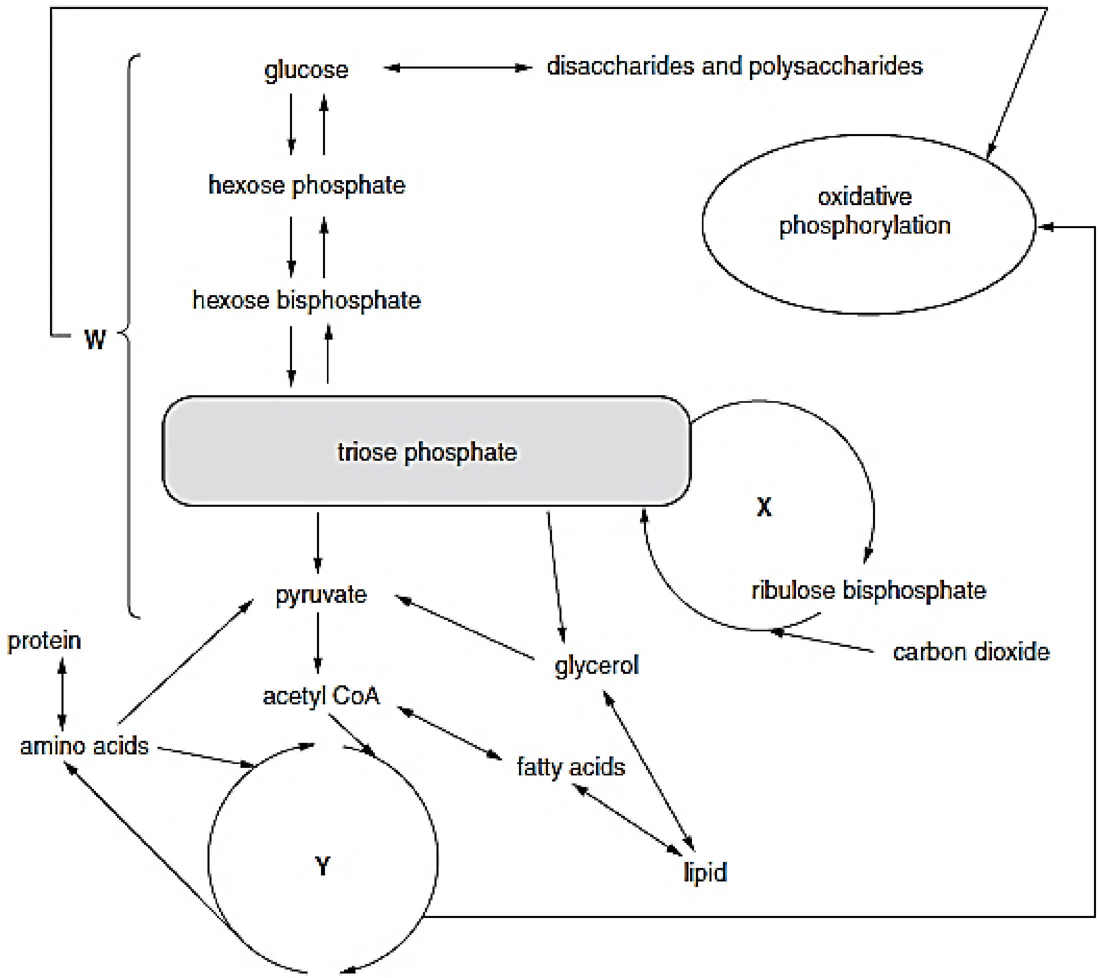
$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O = observed result
E = expected result
v = n–1

Using the equation and the table of χ^2 values, which row correctly describes the result of the χ^2 test?

	number of degrees of freedom (v)	probability	significance of difference between observed and expected results
A	2	< 0.05	significant
B	3	< 0.05	significant
C	2	> 0.05	not significant
D	3	> 0.05	not significant

23 The diagram represents some of the reactions that take place in a leaf cell.



Which statement explains why the three reaction pathways **W**, **X** and **Y** are able to work independently of one another in the same leaf cell?

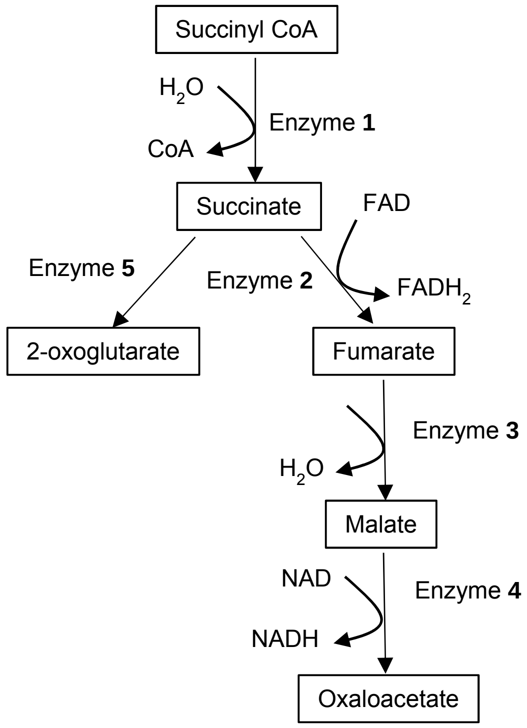
A **W**, **X** and **Y** are separated by membranes, allowing for the maintenance of different conditions for enzymes to function.

B **W**, **X** and **Y** are separated by membranes, allowing for the formation of separate proton gradients to synthesise ATP.

C **X** occurs only in the absence of light unlike both **W** and **Y**.

D Only **X** and **Y** can take place in the absence of oxygen.

24 The diagram shows part of a metabolic pathway in cellular respiration. Each reaction in the pathway is catalysed by a different enzyme numbered **1** to **5**. A high concentration of oxaloacetate inhibits enzyme **2**.



Which will occur in the presence of a high concentration of oxaloacetate?

1 Positive feedback will occur, resulting in the production of more fumarate and malate.

2 Oxidative phosphorylation will cease.

3 Krebs cycle will proceed despite a high concentration of oxaloacetate to form less 2-oxoglutarate.

4 Link reaction will occur with oxidative decarboxylation taking place in the mitochondrial matrix to produce more acetyl-CoA.

A 2 only

B 2 and 3

C 1 and 4

D 1, 3 and 4

25 Trematol is a metabolic poison derived from the white snake root. Cows eating this plant concentrate the poison in their milk. The poison inhibits liver enzymes that convert lactate to other compounds for metabolism.

Which row illustrates the events that occur in an exercising athlete who consumed the trematol-tainted milk?

	lactate accumulation	NAD production	ATP production	pH of blood
A	yes	yes	yes	decreased
B	no	yes	yes	decreased
C	yes	no	no	increased
D	no	no	no	increased

26 Newspaper is made from cellulose. It will only produce a smooth, straight tear when torn in one orientation.

Which structural feature of cellulose accounts for this?

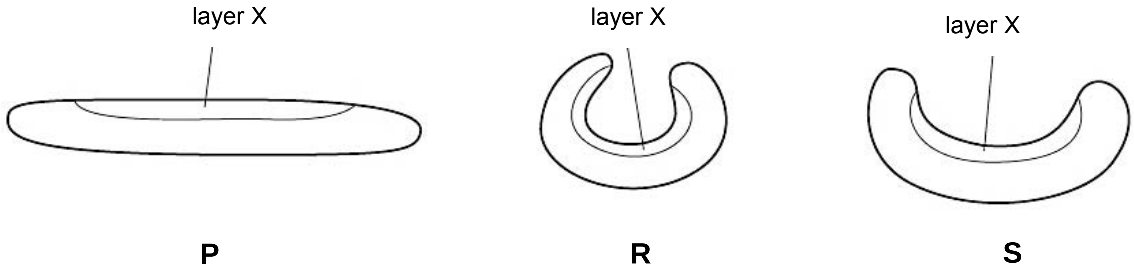
A Cellulose is made up of long, linear chains.

B Cellulose chains have projecting hydroxyl groups for cross-links to form between chains.

C Cellulose is composed only of β-glucose.

D Cellulose chains run parallel to one another.

27 Xerophytes are plant species that are adapted to survive in dry environments. The diagram shows three xerophytic leaves of the same plant species in three different conditions **P**, **R** and **S**.

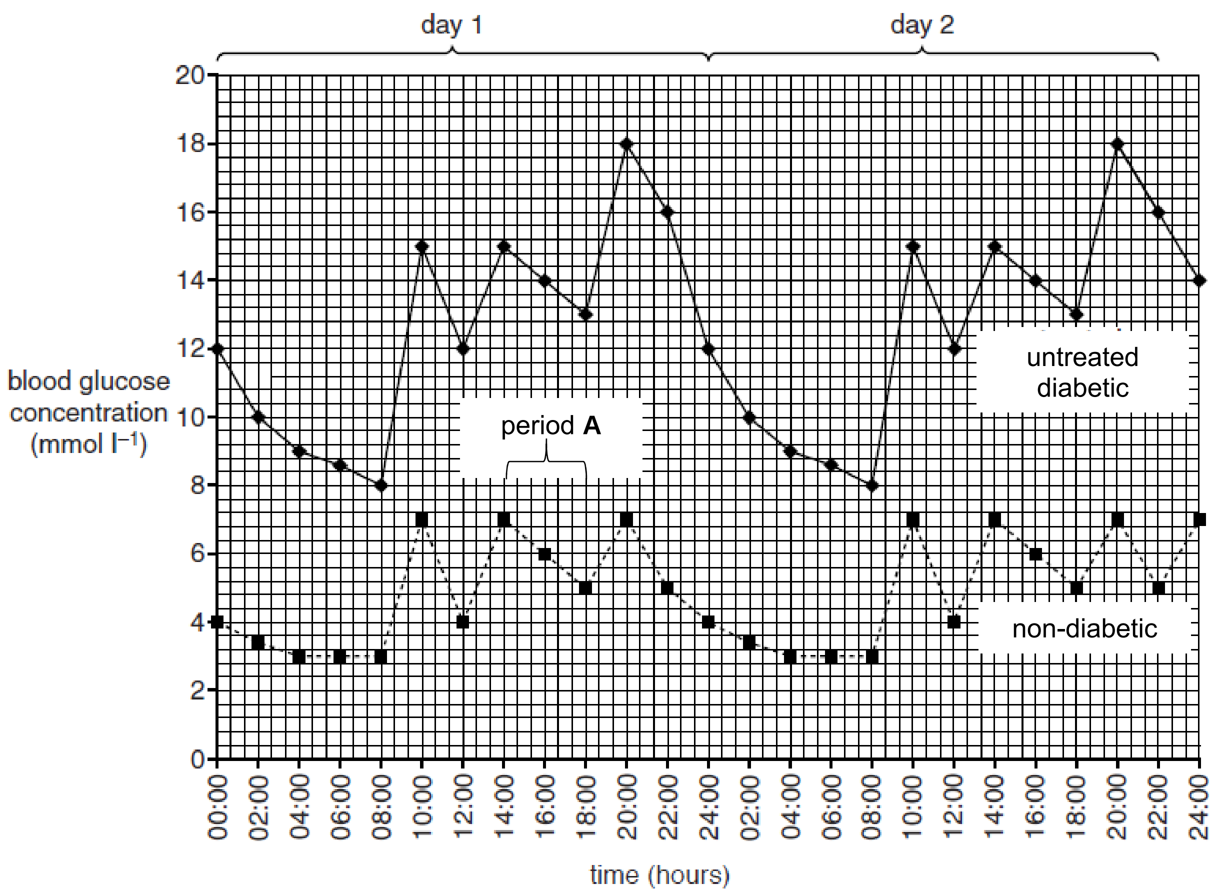


Which description of the water potential of the cells in layer X is correct?

	water potential of cells in layer X		
	P	**R**	**S**
A	less negative than **R** and **S**	more negative than **P** and **S**	more negative than **P**
B	less negative than **S**	more negative than **P**	less negative than **P**
C	more negative than **R**	less negative than **S**	less negative than **R**
D	more negative than R and **S**	less negative than **P**	more negative than **P**

Use the following information to answer **questions 28** and **29**.

The diagram shows the changes in blood glucose concentration over two days in an untreated diabetic and in a non-diabetic.



28 Which statements describing the diabetic could explain the differences between his blood glucose concentration and that of the non-diabetic?

1 Tissues are insensitive to insulin.

2 Less insulin is produced by β cells of the islets of Langerhans in the pancreas.

3 Less glucose uptake by cells occurs.

A 1 and 2

B 1 and 3

C 2 and 3

D 1, 2 and 3

29 Which sequence of cell signalling events occur in liver cells during period **A** to result in the fall in blood glucose concentration?

1 Relay proteins lead to cellular response.
2 Activation of tyrosine kinase activity resulting in autophosphorylation.
3 Receptor aggregation and dimerisation occur.
4 Activated insulin receptors bind to cytoplasmic relay proteins.
5 Insulin binds to insulin receptors.

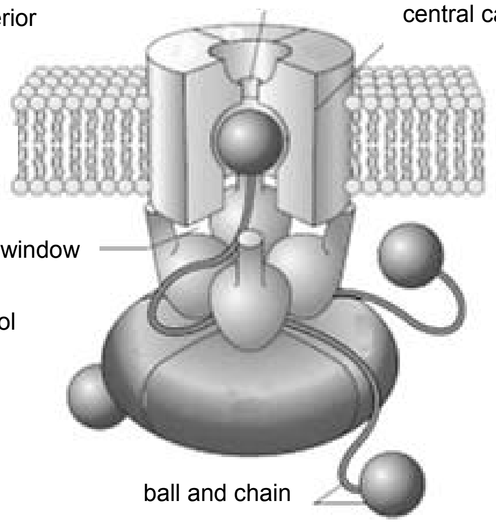
A 3 → 5 → 2 → 1 → 4

B 5 → 2 → 3 → 4 → 1

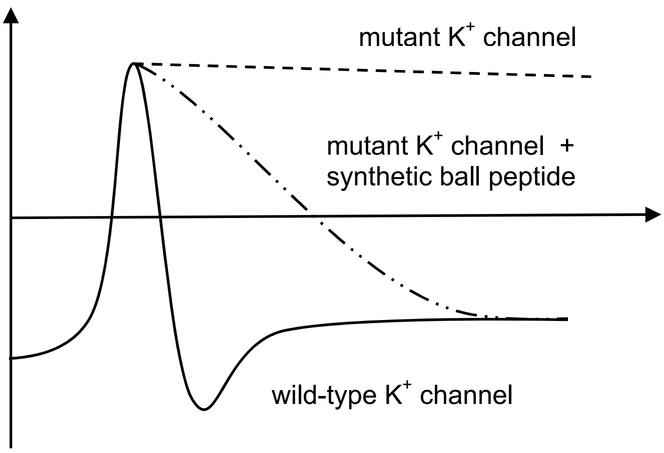
C 3 → 2 → 5 → 1 → 4

D 5 → 3 → 2 → 4 → 1

30 In a voltage-gated K^+ channel, there are four subunits with positively-charged globular balls at their N-termini. In the resting state, the balls are found in the cytosol. Upon depolarisation, one of these balls moves through the "lateral window", blocking the flow of K^+ ions as shown.



An experiment was performed using mutant voltage-gated K^+ channels with mutated globular balls. In another experiment, a chemically-synthesised ball peptide, which mimicked the action of the globular balls of wild-type K^+ channels, was added to the mutant voltage-gated K^+ channels. The graph shows the results.



Which row can be concluded from this experiment?

	stage of action potential	wild-type K^+ channel	mutant K^+ channel	mutant K^+ channel + synthetic ball peptide
A	depolarisation	channel closes, preventing K^+ efflux	channel closes, preventing K^+ efflux	channel opens, allowing K^+ efflux
B	depolarisation	channel opens, allowing K^+ efflux	channel opens, allowing K^+ efflux	channel closes, preventing K^+ efflux
C	repolarisation	channel opens, allowing K^+ efflux	channel closes, preventing K^+ efflux	channel opens, allowing K^+ efflux
D	repolarisation	channel opens, allowing K^+ efflux	channel opens, allowing K^+ efflux	channel closes, preventing K^+ efflux

31 The Amazonian forest today contains a very high diversity of bird species.

- Over the last 2,000,000 years, long periods of dry climate caused this forest to separate into a number of smaller forests.
- Different plant communities developed in each of these smaller forests.
- Each time the climate became wetter again, the smaller forests grew in size and merged to reform the Amazonian forest.

Which statements are true about how the very high diversity of bird species has developed in the Amazonian forest?

1 There was a lack of gene flow between populations.
2 Genetic bottlenecks occurred during the long periods of dry climate.
3 There were different environmental conditions that led to advantageous alleles being selected for and passed on to subsequent generations.
4 Adaptations to the reformed Amazonian forest led to adaptive radiation.

A 1 and 2

B 1 and 3

C 2 and 3

D 3 and 4

32 Early in 2012, biologists announced the discovery of a new lizard species. To claim that the lizards belong to a new species, the biologists must show that the lizards

A look different from one another.

B look different from known species.

C have similar genetic sequences to known species.

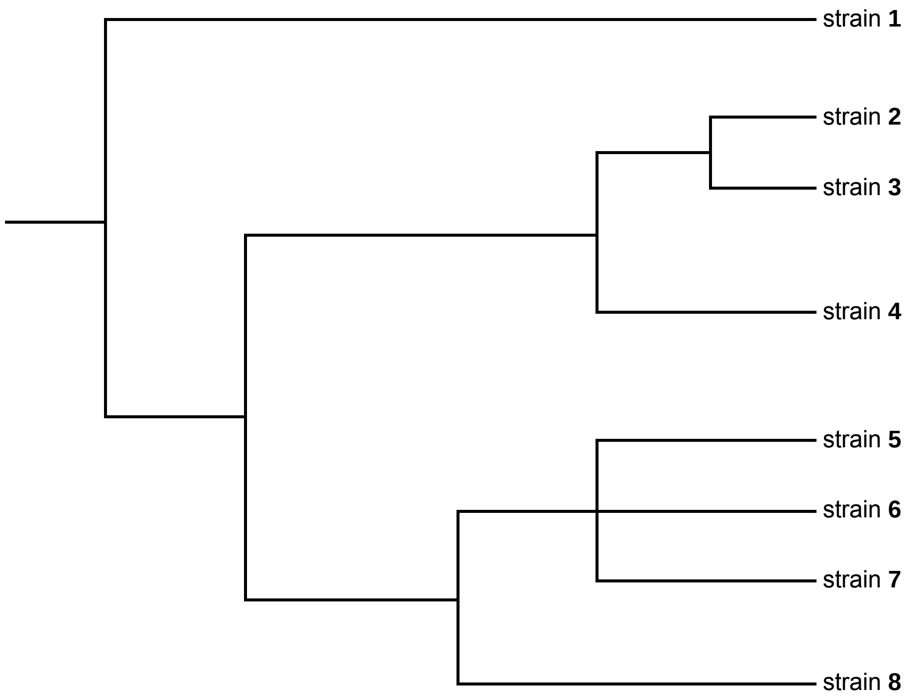
D demonstrate evolutionary relatedness to known species.

33 In humans, Severe Acute Respiratory Syndrome (SARS) is a serious form of pneumonia. SARS is caused by a coronavirus that was first identified in 2003.

Scientists suspected that the virus had been transmitted to humans from some other animal. Testing was completed on several animal species. Strains of the coronavirus similar to those found in humans were identified in different species of horseshoe bats (genus *Rhinolophus*) and palm civets (*Paguma larvata*). Samples were taken from the different sources and the virus's RNA from each sample was sequenced.

The molecular information enabled the scientists to draw an evolutionary tree for different strains of the coronavirus.

The following evolutionary tree was drawn. Strain **7** is found in palm civets, and strains **5** and **6** in humans. All other strains are found in different species of horseshoe bats.



Which suggestion could provide an explanation for the evolution of strains **5**, **6** and **7**?

A Strains **5**, **6** and **7** share the most recent common ancestor, suggesting that strain **7** evolved into strains **5** and **6** over time.

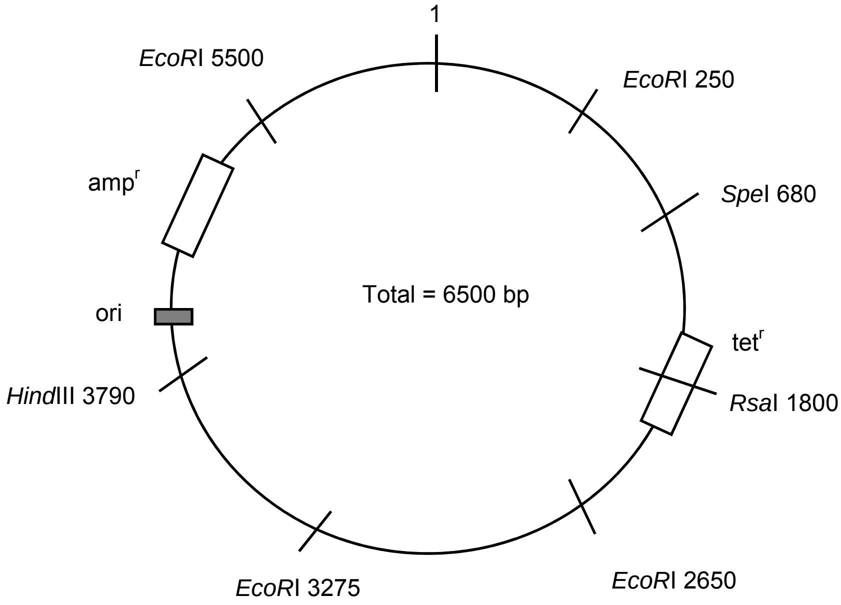
B Strains **5**, **6** and **7** belong to the same genus and species but different families.

C Strains **5**, **6** and **7** possess a shared derived character that distinguishes them from strain **8**.

D Strain **1** is the most distant common ancestor of strains **2** to **8** as it underwent allopatric speciation in the process of evolution.

Use the following information to answer **questions 34** and **35**.

A gene of 1250 base pairs (bp) coding for resistance to leafspot disease in a plant was inserted into a plasmid using the enzyme *Rsa*I. The diagram shows a map of the plasmid.



34 The recombinant plasmid was cut with *Eco*RI, and the restriction fragments were separated by gel electrophoresis.

Which fragments would appear on the gel?

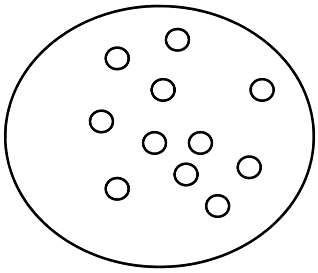
A 1250 bp, 2225 bp, 2400 bp, 3650 bp

B 2650 bp, 3275 bp, 3650 bp, 5500 bp

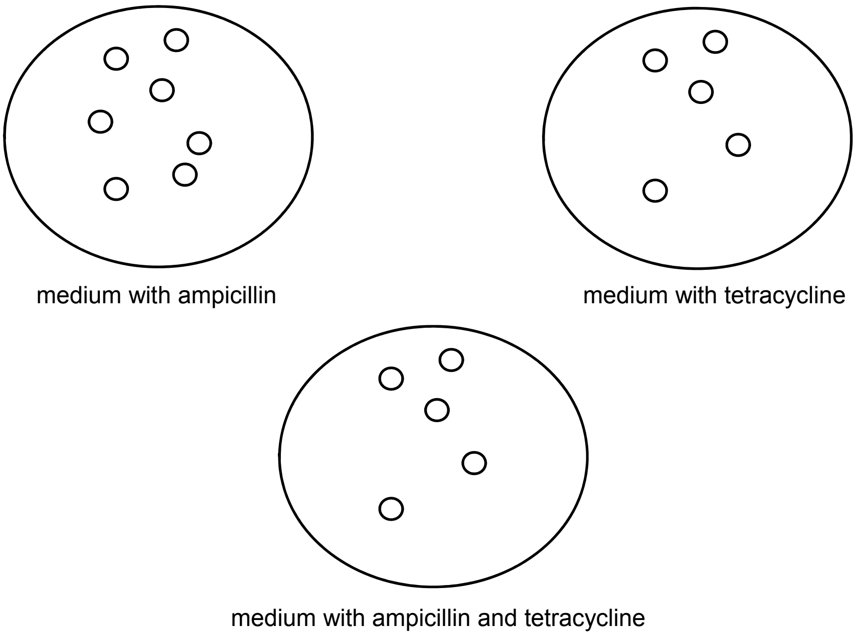
C 250 bp, 2650 bp, 3275 bp, 5500 bp

D 625 bp, 1250 bp, 2225 bp, 3650 bp

35 The recombinant plasmids were transformed into *Escherichia coli* and the bacteria were grown on nutrient medium. The resulting master plate is shown in the diagram.



Transformed cells were selected by replica plating and grown on media containing different antibiotics.



How many colonies in the master plate contain the recombinant plasmid?

A 2

B 4

C 5

D 7

36 A piece of mouse DNA sequence is to be amplified by PCR.

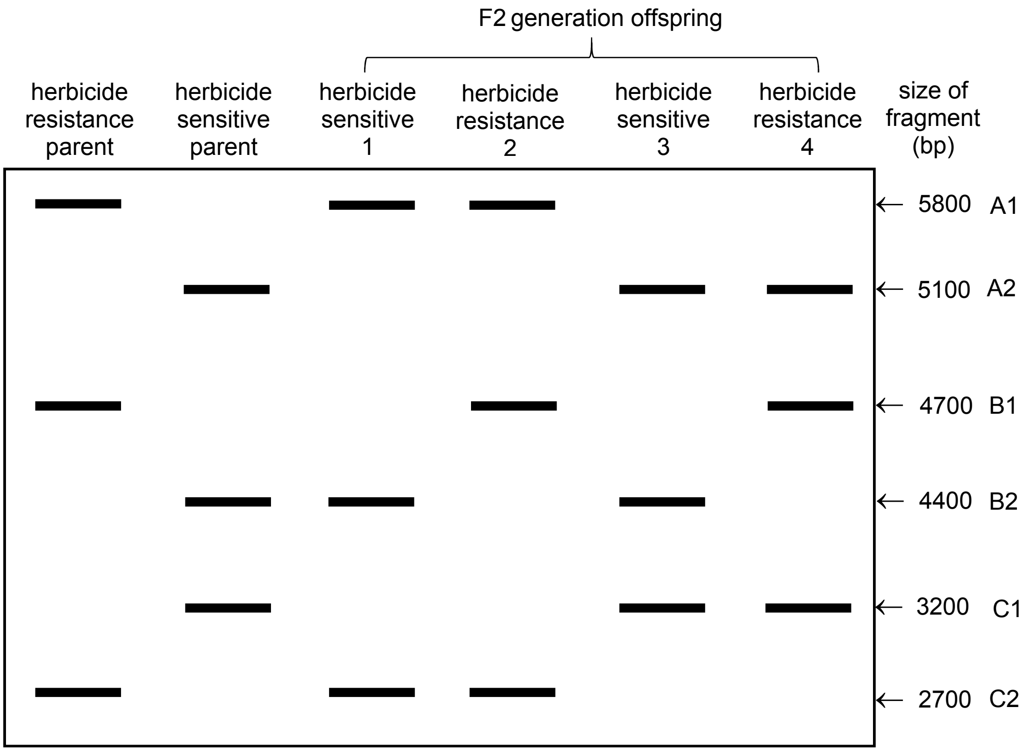
5' AGAGGGCGGT CCGTATCGGC CAATCTGCTC ACCACTAAGC 3'

Which pair of primers should be used?

A	5' AGAGGGCGGT3'	5' CGTTAGTGGT 3'
B	5' CCGTATCGGC 3'	5' TGGTGATTCG 3'
C	5' CCGTATCGGC 3'	5' GCTTAGTGGT 3'
D	5' AGAGGGCGGT3'	5' GCTTAGTGGT 3'

37 Three RFLP loci A, B and C are likely to be linked to the allele for herbicide resistance in maize plants. To locate the resistance allele, the resistance and sensitive varieties are crossed and the F1 offspring are allowed to self-fertilise.

The F2 offspring are then analysed. The results show alleles A1 and A2 on locus A, B1 and B2 on locus B, C1 and C2 on locus C.



Which RFLP allele is the herbicide resistance allele linked to?

A allele A1

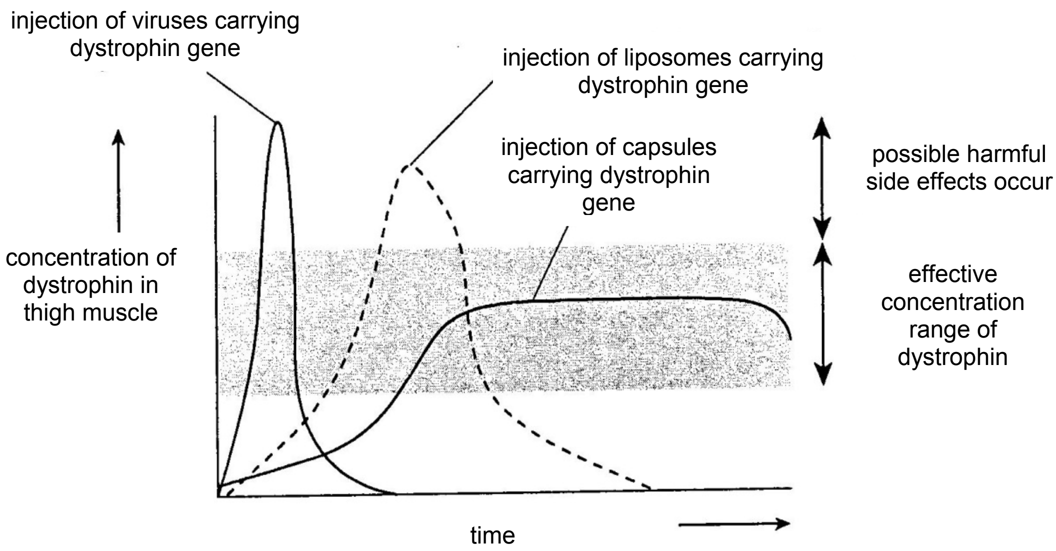
B allele B1

C allele C1

D allele C2

38 Duchenne muscular dystrophy (DMD) is a lethal X-linked human genetic disease caused by the absence of the protein dystrophin in muscle fibres.

Gene therapy experiments were conducted to compare the effectiveness of three different vectors for introducing a corrective gene coding for dystrophin. Each of the three vectors, virus, liposome and capsule, carrying normal copies of the dystrophin gene was injected into the thigh muscle tissue of DMD patients. The results are shown in the graph.



Which is a viable advantage of the capsule vector over the viral and liposome vectors?

A It is more likely to harm the organism.

B Less frequent treatment is needed.

C It takes a shorter time to take effect.

D It takes effect in more target cells.

39 Which method is best suited for the production of virus-free plants?

A embryo culture

B meristem culture

C ovule culture

D anther culture

40 Wild sunflowers are considered to be weeds by many farmers in the USA. They hybridise readily with sunflowers grown as crops. Two experiments involving wild sunflowers gave the following results:

- Up to 42% of the seeds produced by wild sunflowers growing at the margins of a commercial sunflower field were hybrids.
- The hybrid offspring of wild sunflowers and experimental sunflowers produced 50% more seeds than ordinary wild sunflowers. The experimental sunflowers had been genetically engineered to produce insecticide.

Which are potential ecological hazards of growing a genetically engineered crop, such as sunflowers?

1 The hybrid offspring increases the chance of spreading resistance to wild sunflowers, leading to higher chances of invasiveness of the hybrid offspring.

2 Wild sunflowers become resistant because they hybridise readily, becoming more invasive.

3 There is a chance of the hybrid offspring spreading quickly because more seeds are produced, leading to ecological imbalance.

A 1 only

B 2 only

C 1 and 2

D 2 and 3